

Dr. George Close — ML Researcher in Speech & Audio



I am a machine learning researcher specialising in speech and audio. I am currently a Member of Technical Staff on the Audio team at [Zyphra](#), where I help build large-scale text-to-speech and audio foundation models — including the [ZONOS](#) family of TTS systems — working across model architecture, data pipelines and evaluation.

I take ideas from research all the way to production: training and shipping models at scale on large GPU clusters, and grounding them in how humans actually perceive sound. This builds on my PhD in Computer Science from the [University of Sheffield](#) (perceptually-motivated

speech enhancement and quality assessment) and a first-class BSc from [Cardiff University](#), with published work spanning speech enhancement, speech quality prediction and ASR.

Interests

- Text To Speech (TTS) & speech synthesis
- Audio foundation models
- Self-Supervised Speech Representations
- Speech Enhancement / Noise Reduction
- Speech Quality / Intelligibility machine perception and prediction
- Automatic Speech Recognition (ASR)
- Human perception of digital audio
- Neural systems for hearing aids & edge devices
- Deepfake detection / adversarial attacks

Technical Skills

- Large-scale model training on multi-GPU clusters
- Python — PyTorch, SpeechBrain, NVIDIA NeMo, SciPy
- HuggingFace ecosystem / OpenAI API
- Distributed training & data pipelines for audio
- Linux / Bash scripting
- Git / GitHub / GitLab
- C++, Java, SQL, MATLAB

Experience & Qualifications

1. Sep 2025 — Present **Member of Technical Staff @ Zyphra**
San Francisco, CA, USA Research and development on ZONOS, a large-scale text-to-speech model - spanning model architecture, data pipelines and evaluation.
2. Nov 2024 — Sep 2025 **Speech Data Scientist @ ConnexAI**
Manchester, UK Built and deployed production speech-processing systems, including data filtering for in-the-wild speech corpora and non-intrusive speech quality prediction.
3. May 2024 — Aug 2024 [Yamaha Research and Development](#) (Internship)
Hamamatsu, Japan Research internship applying machine learning to audio and music signal processing within an industrial R&D team. Project focused on modeling human perception of music spatiality.

4. Oct 2020 — Jan 2025 **PhD Computer Science + Graduate Teaching Assistant**
University of Sheffield, UK
Thesis: [Perceptually Motivated Speech Enhancement](#) Researched neural speech enhancement guided by human perception — using speech quality metrics and self-supervised representations as loss functions. Authored 10+ first-author papers and taught undergraduate courses as a GTA.
5. Aug 2017 — Aug 2020 **BSc Computer Science (First Class Honours)**
Cardiff University, UK
Thesis: *Majel — Voice control for Command Line Interfaces* Graduated with First Class Honours, with a final-year project building a speech-driven interface for the command line.

Papers & Publications

I am an author on 18 papers, of which 10 I am first author. These have amassed 161 citations with an h-index of 8. Full list on [Google Scholar](#).

- **ZONOS2 Technical Report**

Gabriel Clark, Sofian Mejjoute, Mohamed Osman, [George Close](#), Beren Millidge

2026 Zyphra Technical Report

[arXiv](#)

- **WhiSQA: Non-Intrusive Speech Quality Prediction Using Whisper Encoder Features**

[George Close](#), Kris Hong, Thomas Hain, Stefan Goetze

2025 SPECOM 2025

[arXiv](#) [GitHub](#)

- **Whilter: A Whisper-based Data Filter for "In-the-Wild" Speech Corpora Using Utterance-level Multi-Task Classification**

William Ravenscroft, [George Close](#), Kit Bower-Morris, Jamie Stacey, Dmitry Sityaev, Kris Y. Hong

2025 Interspeech 2025

[arXiv](#) [Interspeech 2025 Dataset](#)

- **Hallucination in Perceptual Metric-Driven Speech Enhancement Networks**

George Close, Thomas Hain, Stefan Goetze

2024 EUSIPCO 2024

[arXiv](#) [EUSIPCO 2024 Listening Test Examples](#)

- **Using Speech Foundational Models in Loss Functions for Hearing Aid Speech Enhancement**

Robert Sutherland, George Close, Thomas Hain, Stefan Goetze, Jon Barker

2024 EUSIPCO 2024

[EUSIPCO 2024](#)

- **Transcription-Free Fine-Tuning of Speech Separation Models for Noisy and Reverberant Multi-Speaker Automatic Speech Recognition**

William Ravenscroft, George Close, Stefan Goetze, Thomas Hain, Mohammad Soleymanpour, Anurag Chowdhury, Mark C. Fuhs

2024 Interspeech 2024

[arXiv](#) [Interspeech 2024](#)

- **Non-Intrusive Speech Intelligibility Prediction for Hearing-Impaired Users using Intermediate ASR Features and Human Memory Models**

Rhiannon Mogridge, George Close, Robert Sutherland, Thomas Hain, Jon Barker, Stefan Goetze, Anton Ragni

2024 ICASSP 2024 🏆 2nd Place — Clarity Prediction Challenge 2

[Technical Report](#) [arXiv](#) [ICASSP 2024](#)

- **Multi-CMGAN+/: Leveraging Multi-Objective Speech Quality Metric Prediction for Speech Enhancement**

George Close, Thomas Hain, Stefan Goetze

2024 ICASSP 2024

[arXiv](#) [ICASSP 2024](#) [GitHub](#)

- **CMGAN+/-: The University of Sheffield CHiME-7 UDASE Challenge Speech Enhancement System**

George Close, William Ravenscroft, Thomas Hain, Stefan Goetze

2023 CHiME-7 UDASE Challenge 🏆 Challenge Entry

[Technical Report Results](#)

- **Non-Intrusive Intelligibility Predictor for Hearing-Impaired Individuals using Self-Supervised Speech Representations**

George Close, Thomas Hain, Stefan Goetze

2023 SPARKS Workshop 2023

[arXiv SPARKS Workshop 2023](#)

- **The Effect of Spoken Language on Speech Enhancement using Self-Supervised Speech Representation Loss Functions**

George Close, Thomas Hain, Stefan Goetze

2023 WASPAA 2023

[arXiv WASPAA 2023](#) [GitHub](#)

- **Perceive and Predict: Self-Supervised Speech Representation Based Loss Functions for Speech Enhancement**

George Close, William Ravenscroft, Thomas Hain, Stefan Goetze

2023 ICASSP 2023

[arXiv ICASSP 2023](#) [Audio Examples](#)

- **PAMGAN+/-: Improving Phase-Aware Speech Enhancement Performance via Expanded Discriminator Training**

George Close, Thomas Hain, Stefan Goetze

2023 154th AES Convention 🏆 Student Technical Paper Award

[AES e-Library](#)

- **Non-Intrusive Speech Intelligibility Metric Prediction for Hearing-Impaired Individuals — Clarity Prediction Challenge 1**

George Close, Samuel Hollands, Thomas Hain, Stefan Goetze

2022 Interspeech 2022

[Interspeech 2022](#)

- **MetricGAN+/-: Increasing Robustness of Noise Reduction on Unseen Data**

George Close, Thomas Hain, Stefan Goetze

2022 EUSIPCO 2022

[arXiv](#) [EUSIPCO 2022](#) [Audio Examples](#)

Talks & Presentations

[AES \[TC-MLAI\] 2022 Talk](#) — "Teaching AI to hear like we do: psychoacoustics in machine learning"